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An open-source application software to determine the preconsolidation pressure of soils in incremental loading oedometer testing: *pySigmaP*

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ABSTRACT

This paper presents *pySigmaP*, an open-source application software developed in Python 3 under the object-oriented paradigm to determine the preconsolidation pressure, σ'_p, of soils in incremental loading oedometer testing. *pySigmaP* was implemented with nine methods: Casagrande, Pacheco Silva, Butterfield, Oikawa, Becker et al., Morin, Onitsuka et al., Wang and Frost, and Boone. Analytical and numerical methods were employed to minimize numerical bias and scale dependency when performing graphical constructions for the determination of σ'_p. Procedure and numerical application of each method were also presented. *pySigmaP* provides a more objective and straightforward analysis to estimate σ'_p of a soil deposit using those nine widely accepted methods.

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Code metadata

Current code version	*0.1.8*
Permanent link to code/repository used of this code version	https://github.com/ElsevierSoftwareX/SOFTX-D-21-00138
Code Ocean compute capsule	https://codeocean.com/capsule/5518935/tree
Legal Code License	*2-Clause BSD*
Code versioning system used	*Git*
Software code languages, tools, and services used	*Python 3.*
Compilation requirements, operating environments	*LaTeX*
& dependencies	*Python 3.6+*
	Python packages: NumPy $\geq$ 1.19.1, SciPy $\geq$ 1.5.0, Matplotlib $\geq$ 3.2.2, Pandas $\geq$ 1.1.1, and mstools $\geq$ 0.1.0
If available Link to developer documentation/manual	pysigmap.readthedocs.io
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1. Motivation and significance

Natural soil deposits experience changes in the *in situ* stresses during their geologic history. Mechanical and chemical processes induce changes in the soil skeleton and fabric. These changes are embedded in the soil structure and are part of the soil stress history. Major changes in the soil structure take place when a sedimentary soil is loaded to a stress state beyond the one that the soil has ever experienced. The preconsolidation pressure in sedimentary soils, σ'_p, determines this transition and can be observed in the compressibility curve. A relatively stiff response at stress levels below σ'_p (i.e., in the recompression range) and a softer response above σ'_p (i.e., in the compression range) are retrieved from compressibility tests. The determination of σ'_p in geotechnical practice is essential not only to predict ground settlements and distortions but also to analyze compressibility soil response. A proper determination of σ'_p is required in geotechnical engineering applications dealing with sedimentary soils, but even in residual soils where the preconsolidation pressure is rather

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associated with yielding states of the soil mass, an adequate determination of that yielding point is key to understand the compressibility soil response.

The preconsolidation pressure is typically obtained from results of incremental loading (IL) oedometer tests [1]. Several researchers have proposed different methods to estimate this mechanical parameter: Casagrande [2], Pacheco Silva [3], Butterfield [4], Oikawa [5], Becker et al. [6], Morin [7], Onitsuka et al. [8], Wang and Frost [9], and Boone [10]. Most of those methods are based on graphical constructions and their procedures usually involve the visual interpretation of recompression and compression zones, which introduces confirmation bias by the analyst when determining the σ'_p-value. [11] demonstrated that the visual approximation of recompression and compression zones in a linear behavior is mainly affected or favored by the absolute scale (i.e., size of the graph) and the relatives scales in the abscissa and ordinate axis, pointing out that an apparent straight line can be made nonlinear and a curve can be straightened by a change of scale. Thus, the coefficient of correlation, R, was recommended to evaluate the "degree" of line straightness since that number is unaffected by any change of scale. Despite this criterion reduces the subjectivity related to the fitting of a straight line by visual inspection, the selection of points to calculate R remains subjective. A similar observation was made by [12], demonstrating that the drawing scale has a visual effect in the curvature of the compressibility curve. The author discussed that the point of maximum curvature in the Casagrande method is scale-dependent, requires vast interpretation, and tends to yield smaller σ'_p-values as the void ratio axis scale increases. Alternatively, the Pacheco Silva method was suggested by the author since it is not scale-dependent. [13] showed that the linear fit of the compression range, also known as normally consolidated (NC) range, affects the determination of σ'_p due to the bias introduced by the analyst. In their study, three authors independently used the Pacheco Silva method to determine σ'_p of three heavily overconsolidated clays with a more rounded compressibility response, yielding a maximum range between σ'_p-values from 165 kPa to 250 kPa (≈50%). This was also evaluated by [10], indicating that the range between σ'_p-values in one example presented by [13] can be even larger. A new interpretation of σ'_p was conducted by the author, resulting in a range from 390 kPa to 640 kPa (≈ 65%).

Despite several methods have been proposed after Casagrande in 1936 introduced the computation of σ'_p of soils in oedometer testing, the availability of software and computational tools to quantify it is scarce. Currently, *soilphysics* developed by [14] in R is to the authors knowledge the only open-source application software available for the calculation of σ'_p, and it is implemented with four methods: Casagrande [2], Pacheco Silva [3], Dias Junior and Pierce [15], and Arvidsson and Keller [16]. On the other hand, the graphical nature of most methods reduces the accuracy and objectivity to determine σ'_p since interpretation is biased by analyst visual skills, engineering judgment, and experience. The goal of this work is to provide a tool to the geotechnical analyst to compute σ'_p via methods commonly used by geotechnical practitioners, improve the objectivity of the methods, and repeatability in the results. This paper and proposed tool are not intended to replace but to help building engineering judgment when determining the preconsolidation or yielding pressure of soils. Points of maximum curvature in the e-log(σ'_v) space, tangent lines, points between load increments, intersection of fitted lines, and line fitting were coded via well-established analytical and numerical methods. Practical recommendations suggested by [6,9,17], and [10] are also taken into account to compute σ'_p. This paper presents the capabilities and implementation of *pySigmaP* to determine σ'_p of soils based on IL odometer testing of nine published well-known geotechnical methods: Casagrande, Pacheco Silva, Butterfield, Oikawa, Becker et al. Morin, Onitsuka et al. Wang and Frost, and Boone.

```
pysigmap/
├─ boone ...  Boone
├─ bilog ...  Bilog
├─ casagrande ...  Casagrande
├─ data ...  Data
├─ energy ...  BeckerEtAl
│              WangAndFrost
└─ pachecosilva ...  PachecoSilva
```

Fig. 1. pySigmaP package structure.

2. Software description

pySigmaP is an application software written in Python 3 under the object-oriented paradigm to determine the preconsolidation pressure of soils in IL oedometer testing. It is published as a free and open-source code under the 2-Clause BSD License and stored on GitHub. The requirements and dependencies of *pySigmaP* to run are listed in the code metadata section.

2.1. Software architecture

Fig. 1 illustrates the modular structure and classes for each module. The software/Python package is divided into six modules: **data.py, casagrande.py, energy.py, bilog.py, pachecosilva.py,** and **boone.py.** The **data.py** module contains the class to load and manage the data of the compressibility soil response. The remaining modules contain the classes and methods to determine σ'_p. Further information regarding each module and basic examples can be found in the software documentation.

3. Software functionalities and illustrative examples

In order to illustrate the methods to compute σ'_p, an IL test from an experimental program conducted by [18] on overconsolidated (OC) Bootlegger Cove clays is used. Table 1 summarizes the test result of the specimen IL-1. In Table 1, C_c- and C_r-values were recalculated from the steepest portion of the compression range and the unload–reload portion, respectively, yielding similar results. Note that the e-value at the beginning of the test (i.e., $\sigma'_v = 0$ and $\varepsilon_a = 0$) is the on-table void ratio, e_0.

3.1. Reading and visualizing data from an IL oedometer test

Data obtained from an IL test must be properly structured to calculate σ'_p. The basic structure for the data is a Pandas DataFrame that can be either directly created with the Python interpreter or loaded from an external file (i.e., .csv or .xls/.xlsx file). Table 2 shows the structure of the data. Units of the effective vertical stress are restricted to kilopascals [kPa], the first row must contain the columns titles or labels (no requirements are needed for the nomenclature), and the second row e_0.

A block-structured script is available online to show the capabilities of *pySigmaP* using the data in Table 1. Block 1 shows how to load the example data using the **Data** class from the **data.py** module. As a second mandatory input of the **Data** class, the σ'_{v0}-value (**sigmaV** attribute) is required. Optional inputs are described in the package documentation. Attributes of the **data** object can be explored running the code line **data.__dict__.** After data loading, visualization of the compressibility curve and σ'_{v0}-, C_c-, and C_r-values can be achieved using the **plot()** method from the **Data** class. Fig. 2a shows the result of the loaded example data given the default input parameters.

Table 1
IL oedometer test on OC Bootlegger Cove clays: Specimen IL-1 [18].

Specimen ID	σ'_{v0} [kPa]	C_c	C_r	Compressibility response			
				σ'_v [kPa]	ε_a [%]	e	Stage
IL-1	75	0.219	0.054	0.00	0.00	0.775	
				6.18	0.87	0.760	
				12.36	1.60	0.747	
				24.81	2.52	0.730	
				49.52	3.72	0.709	Loading
				99.05	5.10	0.685	
				198.19	6.69	0.656	
				396.38	8.92	0.617	
				792.77	11.34	0.574	
				1585.43	14.78	0.513	
				792.77	14.38	0.520	
				396.38	13.69	0.532	
				198.19	12.78	0.548	Unloading
				99.05	11.78	0.566	
				49.52	10.65	0.586	
				99.05	11.01	0.580	
				198.19	11.73	0.567	
				396.38	12.76	0.549	
				792.77	13.89	0.529	Loading
				1585.43	15.51	0.500	
				3170.87	18.78	0.442	
				6341.83	22.50	0.376	
				3170.87	22.33	0.379	
				1585.43	21.62	0.391	
				792.77	20.73	0.407	Unloading
				396.38	19.67	0.426	
				198.19	18.50	0.447	

Note: σ'_{v0}: *in situ* effective vertical stress; C_c: compression index; C_r: recompression index; σ'_v: effective vertical stress; ε_a: axial strain; e: void ratio.

Table 2
Structure of the data to input to the Data class.

	A	B	C
1	Effective vertical stress	Axial strain	Void ratio
2	0.00	0.00	0.775
3	6.18	0.87	0.760
4	12.36	1.60	0.747
5	24.81	2.52	0.730
⋮	⋮	⋮	⋮

3.2. Compression and recompression indices

In order to adequately compute C_c, the IL test must have at least three load increments that define a straight line in the compression range when plotted in logarithmic space, or the maximum load increment must be eight times the estimated σ'_p, as recommended by [1]. Different criteria are available in the **Data** class to compute C_c and C_r. By default, C_c is obtained calculating the steepest slope of the compressibility curve by a cubic spline function. This function interpolates the points of the curve with a piecewise cubic polynomial providing a continuous curve. Thus, the steepest slope is obtained via differentiation. As an alternative, C_c can be calculated by linear regression. The compression range or a portion of it must be specified in the **range2fitCc** parameter of the **compressionIdx()** method. This option allows to control the determination of C_c and can be used for compressibility responses where the compression range changes with stress level (e.g., S-shaped compressibility responses). C_r is computed only by linear regression taking into account three different criteria. The **opt** parameter of the **recompressionIdx()** method must be specified to define the criterion. C_r can be calculated with the first and last point of the first unloading stage (**opt=1**), all the points of the first unloading stage (**opt=2**), or all the points of the unload–reload portion (**opt=3**). The default options are the first criteria. Block 2 shows how to calculate C_c and C_r by different criteria, and Fig. 2b and 2c show the results. Note that the coefficient of determination, R^2, is reported when a linear regression is performed.

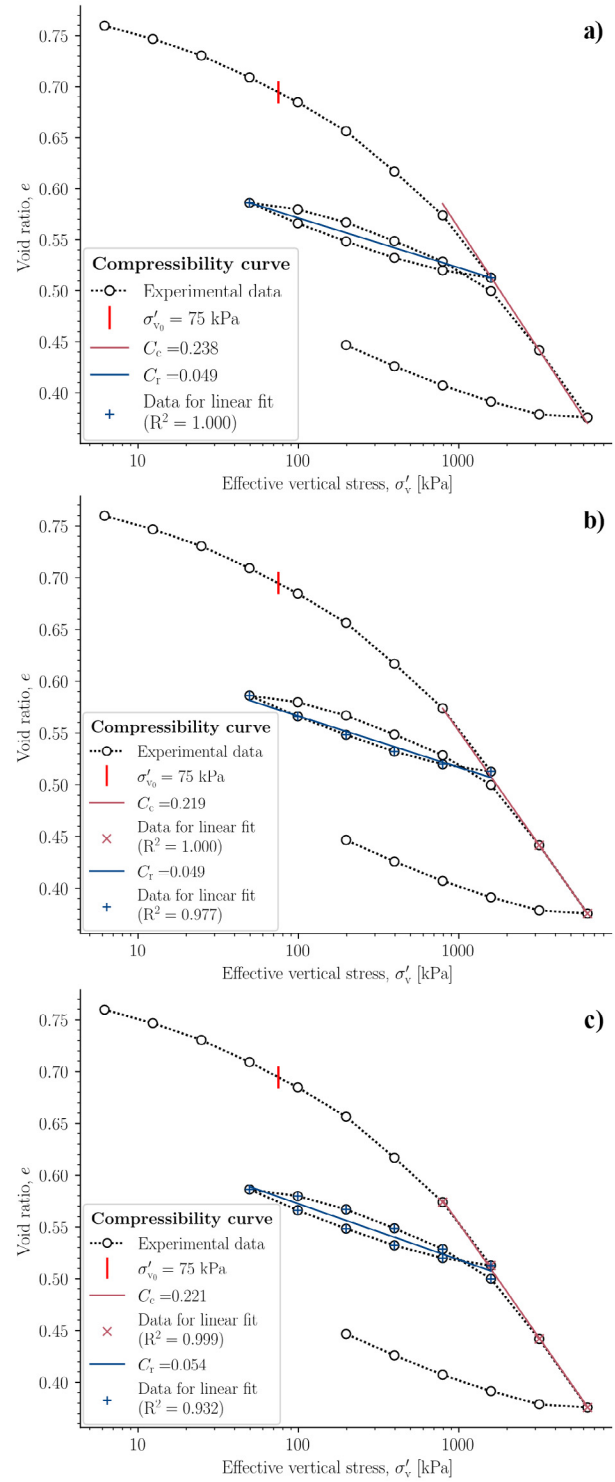

Fig. 2. (a) Compressibility curve plotted given the default input parameters; (b) C_c calculated given the last two points and C_r given all the points of the first unloading stage; and (c) C_c calculated given the last four points and C_r given all the points of the unload–reload portion.

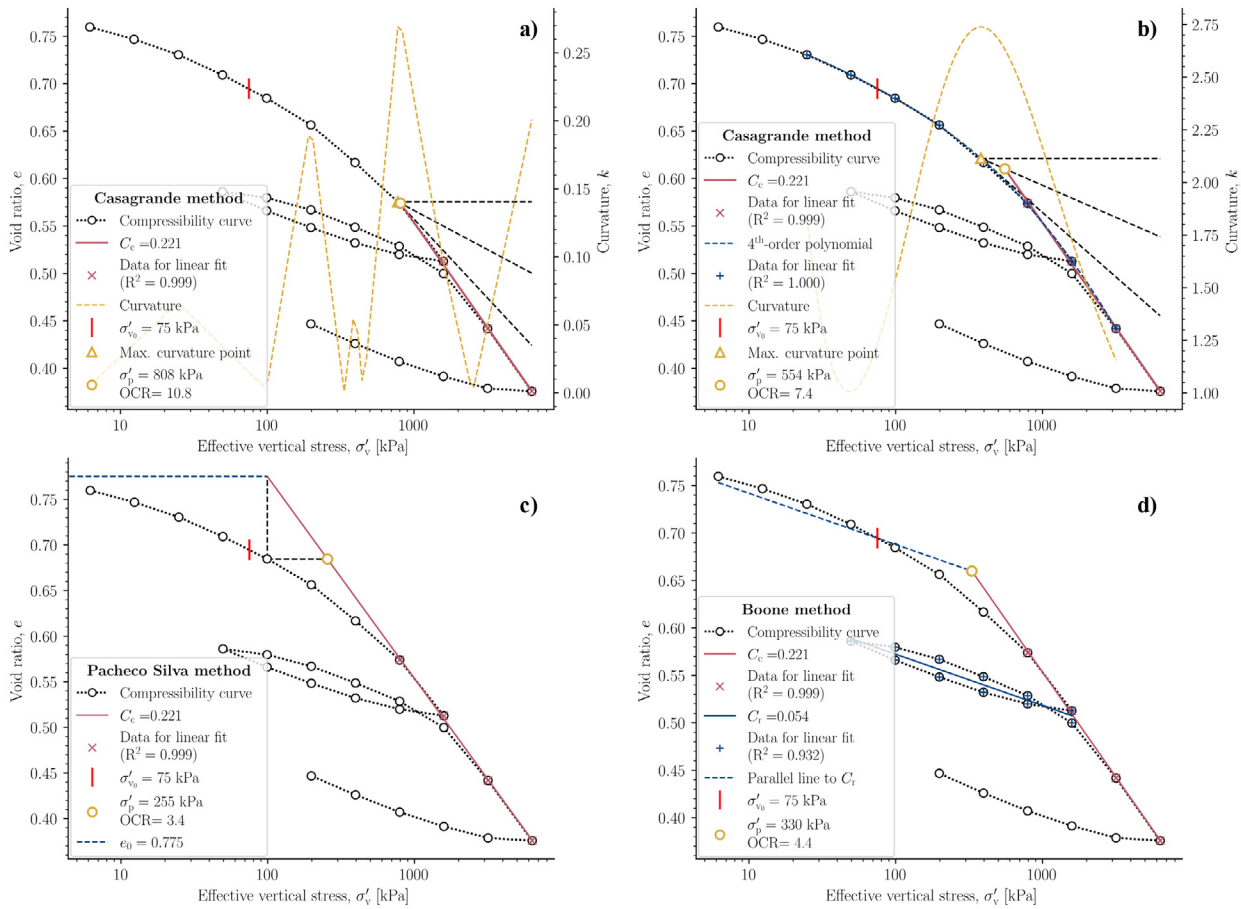

Fig. 3. Computation of σ'_p: (a) Casagrande method (cubic spline function); (b) Casagrande method (FOP); (c) Pacheco Silva method; and (d) Boone method.

3.3. Determination of the preconsolidation pressure

Once the data has been loaded, indices C_c and C_r have been calculated based on the selected method, and once the compressibility curve has been graphically checked, σ'_p can be determined. Each method to compute σ'_p is associated to a class that requires the information of the compressibility curve structured into an object instanced from the **Data** class. When this object is instanced, σ'_p is calculated and graphically represented via the **getSigmaP()** method.

The Casagrande method (**Casagrande** class from the **casagrande.py** module) requires locating the maximum curvature point (MCP). In order to avoid operator visual interpretation and scale dependency biases to locate the MCP, two mathematical approaches proposed by [16,19] were implemented. [19] suggest obtaining the MCP via the mathematical definition of curvature, known as:

$$\kappa = \frac{\left|f''(x)\right|}{\left(1+[f'(x)]^2\right)^{\frac{3}{2}}} \tag{1}$$

where κ is the curvature of the compressibility soil response as a function of the stress level ($x = \log \sigma'_v$). Thus, the MCP is determined by differentiation using the cubic spline function. [16] suggest fitting a fourth-order polynomial (FOP) to the points of the compressibility curve and obtain the MCP with the third derivative of the fitted polynomial equal to zero. Despite the mathematical benefits of constructing a continuous curve by fitting methods, the MCP may be obtained at an unwanted or illogical stress value when the compressibility curve has a rounded shape [16]. A double logarithmic scale of the x-axis is suggested to obtain the MCP and can be applied to the Casagrande method by setting the **loglog** parameter of the **getSigmaP()** method equal to **True**. As an alternative solution, the MCP can be manually introduced by setting the parameter **mcp** of the **getSigmaP()** method with a specific value. Block 3 shows how to calculate σ'_p via the Casagrande method, and Fig. 3a and b show the results. Note that the approach proposed by [16] requires a range to fit the FOP.

The Pacheco Silva method (**PachecoSilva** class from the **pachecosilva.py** module) only requires C_c to calculate σ'_p while Boone method (**Boone** class from the **boone.py** module) require σ'_{v0}, C_c, and C_r. Since these values are previously determined during and after data loading, σ'_p is directly obtained. Block 4 shows how to calculate σ'_p via the Pacheco Silva and Boone methods, and Fig. 3c and d show the results.

The Butterfield, Oikawa and Onitsuka et al. methods (**Bilog** class from the **bilog.py** module) require approximating the recompression and compression ranges with straight lines in bilogarithmic spaces to calculate σ'_p. The recompression range is approximated by linear regression following two criteria: points below σ'_{v0} (**range2fitRR=None**); and selecting points, where the range must be introduced in the **range2fitRR** parameter. Similarly, the compression range is approximated by two criteria: using the cubic spline function to calculate the steepest slope (**range2fitCR=None**); and by linear regression selecting the points. The range is also required in the **range2fitCR** parameter. By default, the first criteria were established to approximate the recompression and compression ranges with straight lines. Bilogarithmic methods are performed by setting the **opt** parameter of the **getSigmaP()** method, **opt=1** for Butterfield method, **opt=2** for Oikawa method, and **opt=3** for Onitsuka et al. method. Block

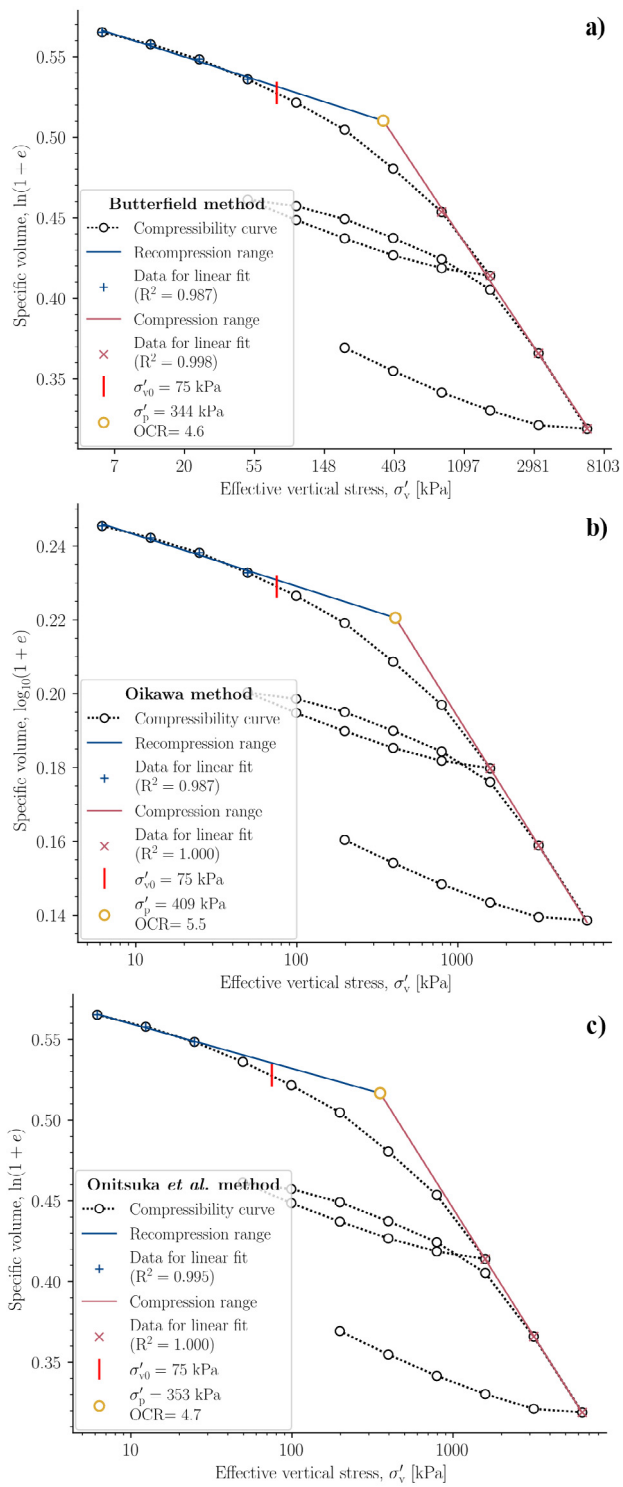

Fig. 4. Computation of σ'_p via the bilogarithmic methods: (a) Butterfield method; (b) Oikawa method; and (c) Onitsuka et al. method.

5 shows how to calculate σ'_p via the bilogarithmic methods, and Fig. 4a, b, and c show the results.

The Becker et al. and Morin methods (**BeckerEtlAl** class from the **energy.py** module) also require approximating the recompression and compression ranges with straight lines, but in this case in arithmetic spaces to calculate σ'_p. The same criteria as in the bilogarithmic methods are used to approximate the recompression and compression ranges. Thus, the **range2fitRR** and **range2fitCR** parameters of the **getSigmaP()** method are employed to select one criterion. [6] recommend to use points below σ'_{v0} to approximate the recompression range in overconsolidated soils (**range2fitRR=None**), and [17] to use the maximum slope instead of the average slope to approximate the compression range (**range2fitCR=None**). Since the Becker et al. and Morin approaches only differ in the mathematical formulation of the work per unit volume for each load increment, the Morin method is applied by setting the **morinFormulation** parameter of the **getSigmaP()** method with **True**.

Finally, the Wang and Frost method (**WangAndFrost** class from the **energy.py** module) only requires approximating the compression range with a straight line to calculate σ'_p. Similar to the previous methods, the **range2fitCR** parameter of the **getSigmaP()** method is employed. It is recommended by [9] to perform a linear regression with the last three points to approximate the compression range. Block 6 shows how to calculate σ'_p via the strain energy methods, and Fig. 5a, b, and c show the results. Note that an inset is incorporated in Fig. 5a and b to magnify the intersection of the recompression and compression ranges. Its width is set as 1.25 times σ'_p, but the zoom value can be changed modifying the **zoom** parameter of the **getSigmaP()** method.

4. Impact

The characteristics and methods coded in *pySigmaP* offer a variety of tools that can be used not only in geotechnical engineering practice but also for research purposes. The methods implemented to determine σ'_p allow to perform comparative studies, evaluate their accuracy, and conduct investigations such as those conducted by [20–22], and [23]. Since different criteria are available to calculate C_c and C_r, their influence on the estimation of primary consolidation settlement for fine-grained saturated soils, or immediate elastic settlement for fine sands can be analyzed. A small python script that operates on *pySigmaP* is required to automate the settlement calculation. The computer-aided calculation of σ'_p provides the geotechnical analyst a tool to examine numerous IL oedometer records in a very short time. The compressibility parameters, C_c, C_r, and σ'_p, are commonly required for the calibration of constitutive soil models implemented in software based on the finite element method (FEM), consequently, *pySigmaP* can serve as a tool to define input values for the numerical modeling of geo-structures. *pySigmaP* is not intended to replace engineering judgment when computing the preconsolidation or yielding pressure of soils, but it is intended to provide a tool to the analyst to evaluate the variables involved in the problem and interpret the results when analyzing sedimentary or residual soils.

It is important to emphasize that regardless of the quality and capabilities of any application software, the quality of the results is as good as the input data. The analyst must account for sample disturbance arising from sampling, improper transportation and storage, handling, and testing procedures, which alter the soil structure obscuring the "break point" in the compressibility response [24–27]. An assessment of the specimen quality via the criteria proposed by [28,29] is suggested before determining the preconsolidation or yielding pressure of soils to obtain a representative value of the *in situ* conditions.

Despite the advantages that offer *pySigmaP*, it has some limitations. Units are restricted to the International System of Units (SI), i.e., unit conversions are not performed during computation of values and graphical outputs. *pySigmaP* was designed to operate with results of IL oedometer tests. The use of results obtained from other testing methods [e.g., constant rate of strain (CRS), controlled gradient (CG), continuous loading (CL),

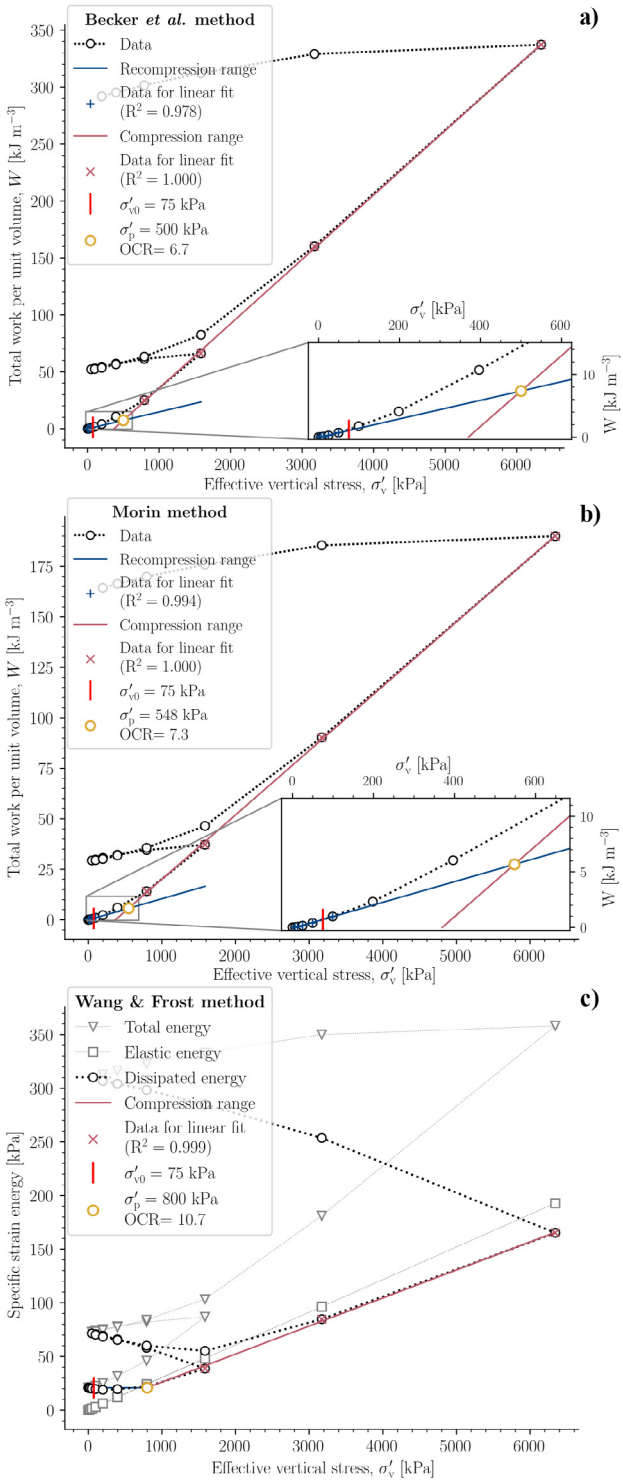

Fig. 5. Computation of σ'_p via the strain energy methods: (a) Becker et al. method; (b) Morin method; and (c) Wang and Frost method.

and K_0-reconsolidation in a triaxial cell (pseudo K_0-TX)] are not recommended. Due to the lack of a graphical interface or a more intuitive environment, interaction with the software is performed via scripting and/or using the Python interpreter.

The analytical and numerical methods implemented in *pySigmaP* improve the objectivity of the methods herein evaluated when computing σ'_p. However, there are aspects that remain subjective and influence the determination of σ'_p, as the selection of load increments to approximate the recompression and compression ranges in straight lines which can be favored or disfavored by the drawing scales. As a consequence, there is an inherent variability when σ'_p is determined. This can be explored with *pySigmaP* as a function of the shape of the compressibility curve, degree of overconsolidation, among others. The R^2-value reported in the graphical output when linear regression is performed can be used as a scale-independent parameter to evaluate the "degree" of line straightness.

5. Summary and conclusions

This paper presented the characteristics and implementation of an open-source application software designed to determine the preconsolidation pressure of soils in incremental loading oedometer testing by nine methods commonly used in geotechnical engineering practice. *pySigmaP* provides a more objective and straightforward analysis to estimate σ'_p of a soil deposit because its calculation is guided by analytical and numerical methods minimizing bias by the scale dependency and analyst visual skills, engineering judgment, and experience. Since different methods, criteria, and recommendations by different researchers are implemented, *pySigmaP* is suitable to perform comparative studies, evaluate the accuracy of the methods to calculate σ'_p, analyze the influence of the methods to calculate C_c and C_r on the estimation of ground settlement, and examine numerous IL oedometer records in a very short time. *pySigmaP* is not intended to replace but to help building engineering judgment providing a tool to the geotechnical analyst to evaluate and interpret compressibility soil response. This application software is published as a free and open-source code and be found online at GitHub.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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